

## Sum of N natural no.

$$1+2+3+4+5=15$$

Approach1 : Mathematical formula=  $N*(N+1)/2$

$$5*(5+1)/2=15$$

Approach2 : Loop based approach

- Find **sum** --> addition --> assign a **sum** variable equals to 0.
- Find **product** --> multiply --> assign a **product** variable equals to 1.

### Dry Run :

```
sum=0
i=1
```

1 2 3 4 5

```
while(i<=N):
    sum =sum+i
```

```
i+=1
```

```
print(sum)
```

| Iteration 1:                             | Iteration 2:                             | Iteration 3:                             | Iteration 4:                             | Iteration 5:                             | Iteration 6:                             |
|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| 1<=5 <input checked="" type="checkbox"/> | 2<=5 <input checked="" type="checkbox"/> | 3<=5 <input checked="" type="checkbox"/> | 4<=5 <input checked="" type="checkbox"/> | 5<=5 <input checked="" type="checkbox"/> | 6<=5 <input checked="" type="checkbox"/> |
| sum = sum + 1                            | sum = sum + 2                            | sum = sum + 3                            | sum = sum + 4                            | sum = sum + 5                            | <b>LOOP STOP</b>                         |
| Sum =1                                   | Sum =3                                   | Sum =6                                   | Sum =10                                  | Sum =15                                  |                                          |
| i=2                                      | i=3                                      | i=4                                      | i=5                                      | i=6                                      |                                          |